CTP0007 Issue H Calculation Procedure

This document describes the calculation procedure implemented in `coupling_calculations.py` for the CTP0007 Issue H desktop application. Units are N, mm, MPa, and Nm unless otherwise noted. MPa is treated as N/mm2.

1. Input Selection

The screw type and size select a screw record from the embedded CTP 0007 lookup data. For `SPECIAL` screws, the user-entered geometry is used where applicable.

Material yield strength is selected from the material database unless the editable `TYS MPa` field is changed by the user.

Tightening input priority is:

1. User-entered tightening torque.
2. User-entered percent of tensile yield strength, `% TYS`.
3. Standard tightening torque from the selected screw type, size, and material.

If both tightening torque and `% TYS` are entered, the calculation is rejected.

2. Geometry

Definitions:

- `d` = thread major diameter
- `p` = thread pitch
- `ds` = shank diameter
- `dg` = groove diameter
- `dc` = contact diameter
- `Sy` = bolt tensile yield strength
- `n` = screw count
- `PCD` = pitch circle diameter

Thread pitch diameter:

```
dp = d - 0.6495 * p
```

Thread root diameter:

```
dr = d - 1.2268 * p
```

Tensile stress diameter:

```
dt = (dp + dr) / 2
```

Tensile stress area:

```
As = pi * ((dp + dr) / 4)^2
```

Shear and bending diameter:

```
dsb = dt    when shear plane is Thread
dsb = ds    when shear plane is Shank
```

Contact diameter:

```
dc = special contact diameter    for SPECIAL screws
dc = special contact diameter    when Nut contact is Special and value > 0
dc = standard record contact diameter otherwise
```

Average annular nut/contact radius:

```
Ro = dc / 2
Ri = dp / 2
Rn = (2 / 3) * ((Ro^3 - Ri^3) / (Ro^2 - Ri^2))
```

The contact diameter must be greater than the pitch diameter.

3. Effective Lever Arm

Effective shear/bending lever arm:

```
Leff = 0.05 mm                          for Stripper bolt
Leff = 0.15 * pack_thickness_mm         for Drive bolt
Leff = 0.15 * pack_thickness_mm         for Shim
```

Drive bolt and Shim require a positive pack thickness. Stripper bolt uses `N/A` pack thickness.

4. Tightening, Pretension, and Friction Torque

Friction definitions:

- `mu_sn` = screw/nut friction coefficient
- `mu_np` = nut/part friction coefficient
- `mu_pp` = part/part friction coefficient
- `Tt` = selected tightening torque
- `Fv` = axial pretension

Axial pretension from tightening torque:

```
Fv = 1000 * Tt / (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn)
```

Axial pretension from `% TYS`:

```
Fv = (%TYS / 100) * As * Sy
```

Reported tightening torque when `% TYS` is used:

```
Tt_reported = (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn) * Fv / 1000
```

Preload percent of tensile yield strength:

```
preload_%TYS = Fv / (As * Sy) * 100
```

Nut/part friction torque component:

```
T_np = mu_np * Rn * Fv / 1000
```

Torsional tightening torque in the screw:

```
T_torsion = (0.16 * p + 0.583 * mu_sn * dp) * Fv / 1000
```

Flange friction torque capacity:

```
T_friction = Fv * n * mu_pp * PCD / 2 / 1000
```

Friction torque ratio is displayed as information:

```
ratio_% = T_friction / T_duty * 100
```

Friction torque lower than duty torque is not a warning by itself. Any uncovered torque is carried forward into the bolt shear/bending calculation.

5. Duty Torque Cases

The app evaluates three duty torque cases:

- Continuous
- Peak
- Momentary

For each case:

```
T_residual = max(0, T_duty - T_friction)
V_total = 2000 * T_residual / (PCD * n)
```

`V_total` is the shear load per joint/bolt station.

If a sleeve is present and sleeve OD is greater than the thread major diameter:

```
V_sleeve = V_total * (OD^2 - d^2) / OD^2
V_bolt = V_total - V_sleeve
```

If no sleeve is present:

```
V_sleeve = 0
V_bolt = V_total
```

6. Sleeve Stress and Safety Factor

Sleeve check is only numeric when `sleeve OD > d`. Otherwise the result is `No Sleeve`.

Sleeve bending stress:

```
I_sleeve = pi / 64 * (OD^4 - d^4)
sigma_b_sleeve = V_sleeve * Leff * OD / 2 / I_sleeve
```

Sleeve shear stress, as implemented:

```
tau_sleeve = V_total / (pi / 4 * OD^2)
```

Sleeve Von Mises stress:

```
VM_sleeve = sqrt(sigma_b_sleeve^2 + 3 * tau_sleeve^2)
```

Sleeve yield safety factor:

```
SF_sleeve = sleeve_yield / VM_sleeve
```

Sleeve material options:

```
GMC 0401  = 245 MPa
GMC 0336  = 650 MPa
GMC0433   = 800 MPa
```

When sleeve OD is `N/A`, sleeve material and sleeve yield are `N/A`.

7. Bolt Shear/Bending Area Stress and Safety Factor

Bolt axial stress:

```
sigma_axial = 4 * Fv / (pi * dsb^2)
```

Bolt bending stress:

```
I_bolt = pi / 64 * dsb^4
sigma_b_bolt = V_bolt * Leff * dsb / 2 / I_bolt
```

Bolt shear stress:

```
tau_bolt = V_bolt / (pi / 4 * dsb^2)
```

Bolt maximum Von Mises stress:

```
VM_bolt = sqrt((sigma_axial + sigma_b_bolt)^2 + 3 * tau_bolt^2)
```

Bolt yield safety factor:

```
SF_bolt = Sy / VM_bolt
```

The table column `Maxi Stress (VM) MPa` reports `VM_bolt`.

8. Assembly Stress in Groove Area

If `dg <= 0`, the result is `No groove`.

Groove axial stress:

```
sigma_groove = 4 * Fv / (pi * dg^2)
```

Groove twisting stress:

```
tau_groove = 16 * T_torsion * 1000 / (pi * dg^3)
```

Groove Von Mises stress:

```
VM_groove = sqrt(sigma_groove^2 + 3 * tau_groove^2)
```

Groove yield safety factor:

```
SF_groove = Sy / VM_groove
```

9. Assembly Stress in Thread Root Area

Thread-root axial stress:

```
sigma_thread = Fv / As
```

Thread-root twisting stress:

```
tau_thread = T_torsion * 1000 * dr / 2 / (pi / 32 * dr^4)
```

Thread-root Von Mises stress:

```
VM_thread = sqrt(sigma_thread^2 + 3 * tau_thread^2)
```

Thread-root yield safety factor:

```
SF_thread_root = Sy / VM_thread
```

10. Thread Pull-Out Stress

Thread engagement:

```
L = user-entered engagement
L = 1.2 * d    when thread engagement mode is Thread/default
```

Thread pull-out area:

```
A_pullout = pi / 2 * dp * L
```

Engaged thread quantity:

```
N_engaged = L / p
```

First-thread load factor for `Emuge` or `Prevailing` screw/nut friction presets:

```
K = max(1,
        0.001762 * N_engaged^3
```

```
      - 0.028314 * N_engaged^2
      + 0.182016 * N_engaged
      + 0.720405)
```

First-thread load factor for other screw/nut friction presets:

```
K = max(1,
        0.002317 * N_engaged^3
      - 0.039254 * N_engaged^2
      + 0.416953 * N_engaged
      + 0.436158)
```

Thread pull-out stress:

```
sigma_pullout = Fv / A_pullout * K
```

The tapped-hole yield strength field is optional. If entered, tapped-hole yield must be greater than or equal to thread pull-out stress.

11. Minimum Safety Factor

The reported minimum safety factor is the minimum of all numeric safety factors:

```
minimum_SF = min(
    all bolt shear/bending SF values,
    groove SF when groove exists,
    thread-root SF,
    all sleeve SF values when sleeve exists
)
```

12. Checking Criteria

12.1 API 671 5th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

```
Continuous torque: SF >= 1.50
Peak torque:       SF >= 1.15
Momentary torque:  SF >= 1.00
```

12.2 API 671 4th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

```
Continuous torque: SF >= 1.25
Peak torque:       SF >= 1.15
Momentary torque:  SF >= 1.00
```

12.3 Groove Area

```
SF_groove >= 1.10
```

12.4 Thread Root Area

```
SF_thread_root >= 1.00 required
SF_thread_root >= 1.10 preferred
```

The preferred value is used because residual torque can reduce by about 20 percent upon loading.

12.5 Tightening Preload Percent

```
Stripper bolt: preload_%TYS <= 60
Shim:          preload_%TYS <= 60
Drive bolt:    preload_%TYS <= 75
```

12.6 Thread Pull-Out

```
tapped_hole_yield >= sigma_pullout
```

If tapped-hole yield is blank or zero, the app reports the pull-out stress with a note rather than a pass/fail result.

13. Input Validation

The calculation rejects invalid input before solving:

- `PCD > 0`
- `screw_count >= 1`
- `thread > 0`
- `pitch > 0`
- checking standard must be recognized
- joint type must be recognized
- pack thickness must be zero or positive
- Drive bolt and Shim require positive pack thickness
- contact diameter must be greater than zero
- tightening torque and `% TYS` cannot both be entered
- screw/nut, nut/part, and part/part friction coefficients must be greater than zero
- sleeve OD must be zero or positive
- tapped-hole yield must be zero or positive
- lever arm must be zero or positive

14. Legacy Quick Friction Model Appendix

The module retains an older quick friction fastener model for compatibility and tests. It is not the main CTP0007 Issue H GUI workflow.

Effective friction radius:

```
r_eff = (inner_radius + outer_radius) / 2
```

when the radius model is uniform wear, otherwise:

```
r_eff = (2 / 3) * ((outer_radius^3 - inner_radius^3) /
                   (outer_radius^2 - inner_radius^2))
```

Design torque:

```
T_steady = transmitted_torque * service_factor
T_design = max(T_steady, cyclic_torque, transient_torque)
```

Residual pretension after preload loss:

```
F_residual = F_initial * (1 - preload_loss_percent / 100)
```

Slip torque capacity:

```
T_slip = mu * (F_residual * bolt_count) * r_eff * friction_interfaces / 1000
SF_slip = T_slip / T_design
```

Required clamp load:

```
F_required_total = T_design * 1000 / (mu * r_eff * friction_interfaces)
F_required_per_bolt = F_required_total / bolt_count
F_required_initial = F_required_per_bolt / (1 - preload_loss_percent / 100)
```

Bolt yield and proof loads:

```
F_yield = yield_strength * tensile_area
F_proof = proof_strength * tensile_area
```

Bolt/joint stiffness split:

```
phi = k_bolt / (k_bolt + k_joint)
F_bolt_service = F_residual + phi * F_axial
F_residual_service = F_residual - (1 - phi) * F_axial
```

Yield utilization values:

```
assembly_yield_utilization = F_initial / F_yield
service_yield_utilization = F_bolt_service / F_yield
required_preload_yield_utilization = F_required_initial / F_yield
```

Legacy quick model warnings include low service factor, transient/cyclic torque governing, slip capacity below demand, preload/yield utilization above the selected limit, preload above proof load, and joint separation.